



WattExchange

**AI-matched, Solana settled, P2P
Solar Energy Surplus Trading
Platform**

**By Team 3
SDG: 7 & 11**

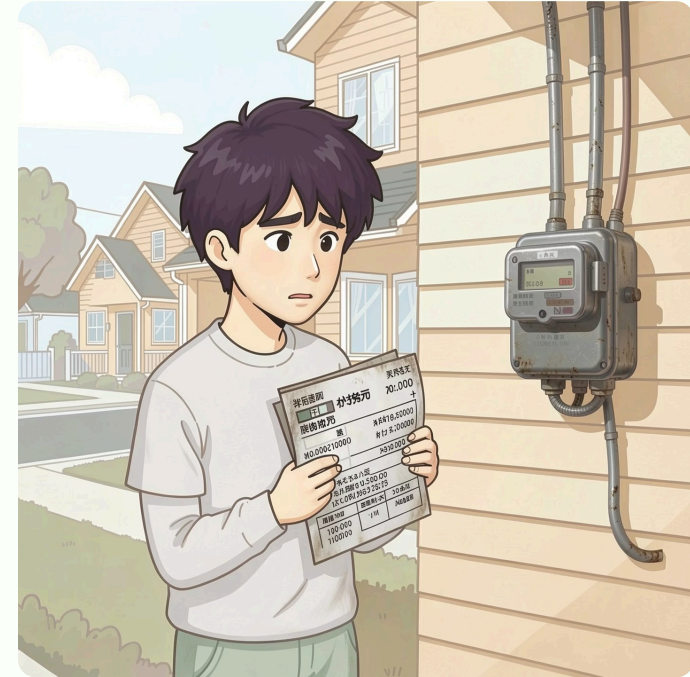
Meet Sarah and Alex

☐ **Grid rate = \$0.12 / kWh | Credit received for selling surplus back to the grid (FIT) = \$0.08 / kWh**



Sarah ☀️

Her solar panels generate extra electricity, especially in summer and during peak afternoon hours. She has surplus energy to sell back to the grid.

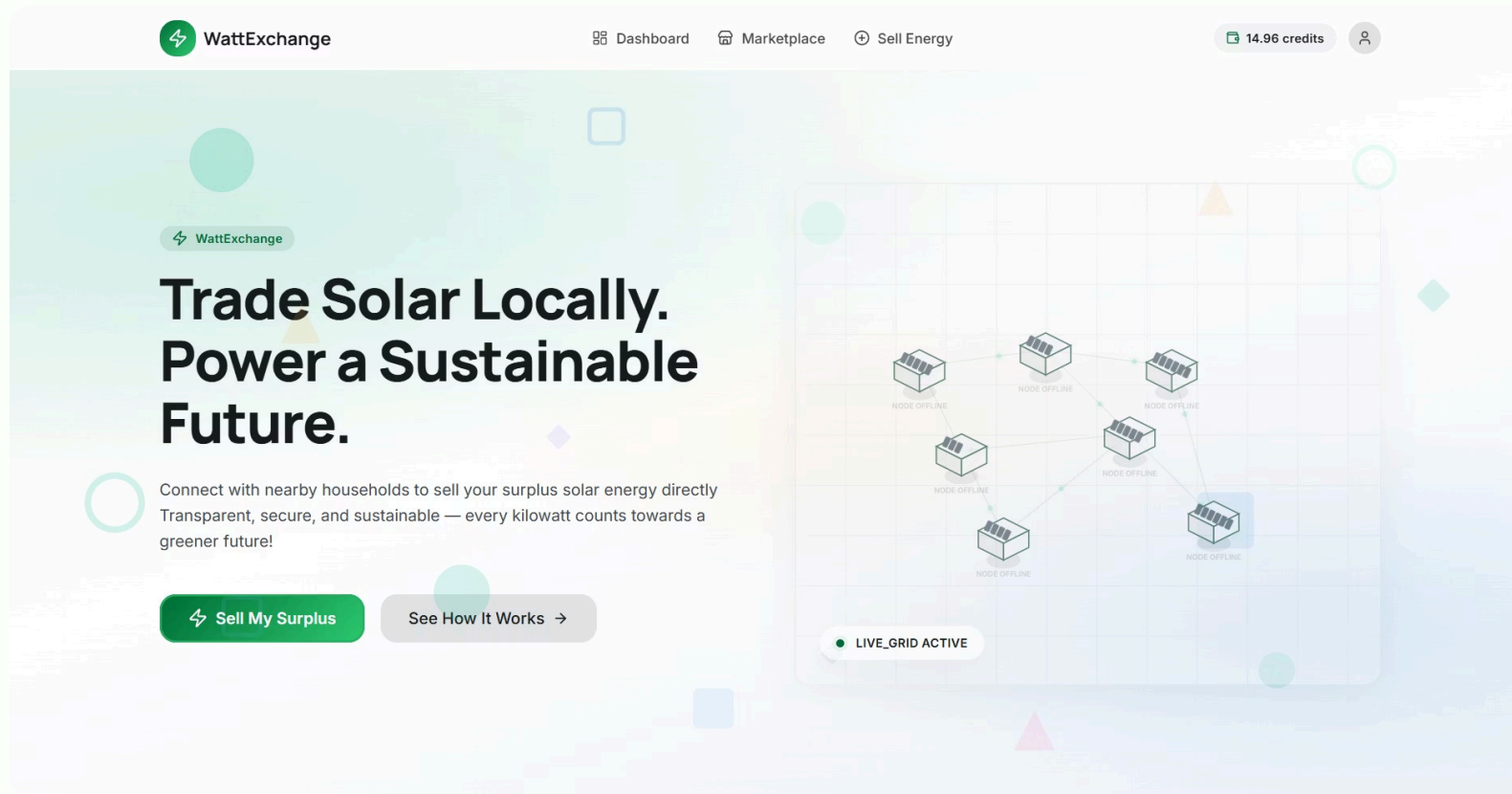


Alex 🏠

He lives in the same neighborhood, does not have solar panels, so he relies on the grid and pays high electricity bills.

Presenting, WattExchange

A P2P Solar Energy Surplus Trading Platform !



How It Works



Surplus Listing

LSTM model auto-lists surplus or prosumers manually list excess kW with a custom price.



AI-Matched Listings

Buyers can browse listings by proximity, price, and reliability. Best match wins, automatically.



Solana Settlement

USDC locked in escrow via smart contract.
Released automatically on delivery confirmation.
~400ms finality.



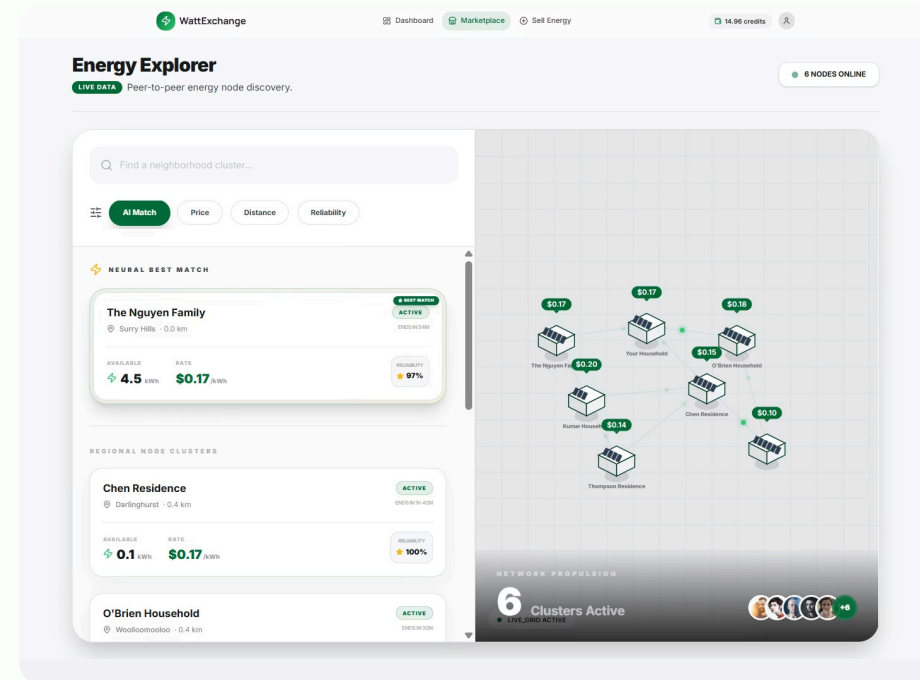
Grid Fallback

Unsold surplus returns to the grid. Nothing wasted.
Platform earns a 1- 2% fee on completed trades.

Key Features

Marketplace and Pricing

- Our matching engine – **running every 15 minutes** – scores every available listing by proximity, price, and reliability using a weighted algorithm and the **Haversine formula** for geolocation. The best match wins.
- **View all active listings** with distance, available kW, and price/kWh
- Optional filters for buyers

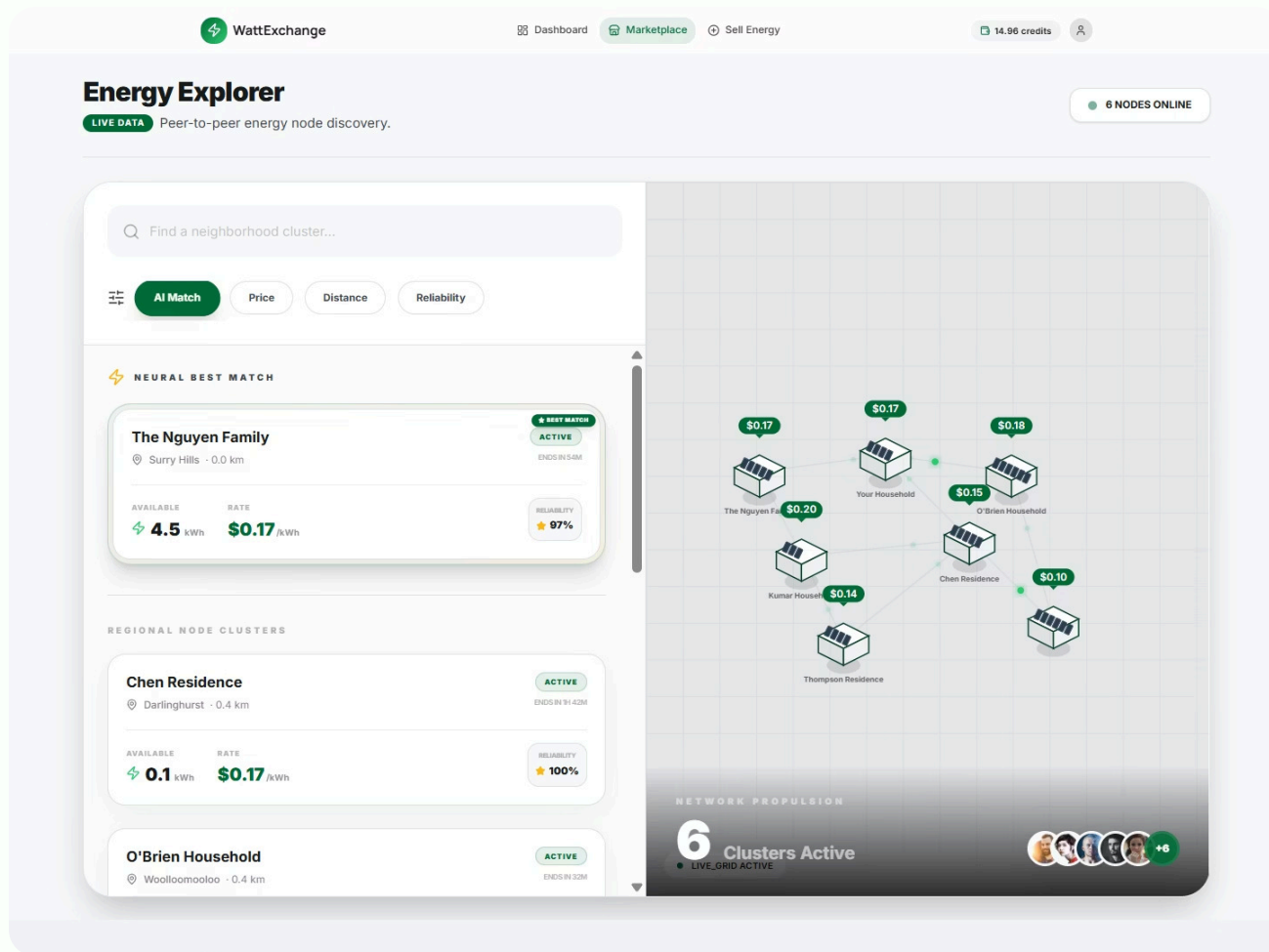


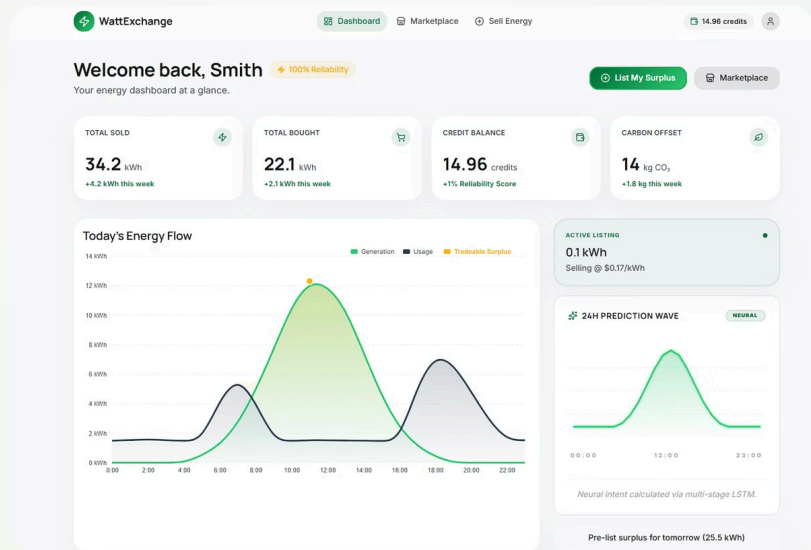
Key Features



Smart Energy System

- Geolocation based matching
- Listings update dynamically as trades happen
- Surplus expires end of the day and automatically returns to the grid





Key Features

AI Forecasting



- Showing prosumers "you'll have 8.2 kWh tomorrow, list it now?"
- LSTM model predicts next day's surplus based on historical consumption and generation readings

Key Features

 My Wallet

4Ugui...yCaW

SOL Balance
4.9390
SOL

Credits
15
WattCredits

 Trade History

4oWdVJnT...
Slot 453748117 · To: 9P643o3W...

0.02008 SOL

3MZqu1gi...
Slot 453354085 · To: 9P643o3W...

0.02008 SOL

28vTh7JV...
Slot 453302093 · To: 9P643o3W...

0.02008 SOL

3mx5zyjx...
Slot 453301851 · To: 9P643o3W...

0.02008 SOL

4Qk1sxJz...
Slot 453301687 · To: 9P643o3W...

0.02008 SOL

Recent Trades					
Counterparty	Energy	Cost	Status	TX	Date
The Nguyen Family SELLER	0.1 kWh	\$0.02	settled		6 Apr
The Nguyen Family SELLER	0.1 kWh	\$0.02	settled		4 Apr
The Nguyen Family SELLER	3 kWh	\$0.51	settled		4 Apr
O'Brien Household SELLER	2.1 kWh	\$0.38	settled		4 Apr
Chen Residence SELLER	4 kWh	\$0.60	settled		4 Apr

Blockchain Layer: Solana



- Phantom wallet stores energy credits
- View TX signature corresponding to each trade
- Transparent and secure transactions
- Trade history showing past transactions

Technical Overview

Frontend & Backend

- React + NodeJs and TypeScript

Blockchain Integration

- We use Solana Devnet, which is a test network with fake SOL, so no real money is used
- Phantom Wallet Integration for energy credits (SOL)

Data Simulation & Matching

- LSTM AI model: Energy meter readings, surplus, and geolocation-based matching are formula-driven, not hard-coded, to mimic real-world behaviour
- Best-Match Filter: Buyers see top listings based on a weighted score of price and proximity, normalized to provide a combined ranking.
- Marketplace Matching: Uses browser geolocation and the Haversine formula to calculate distances between buyers and sellers.



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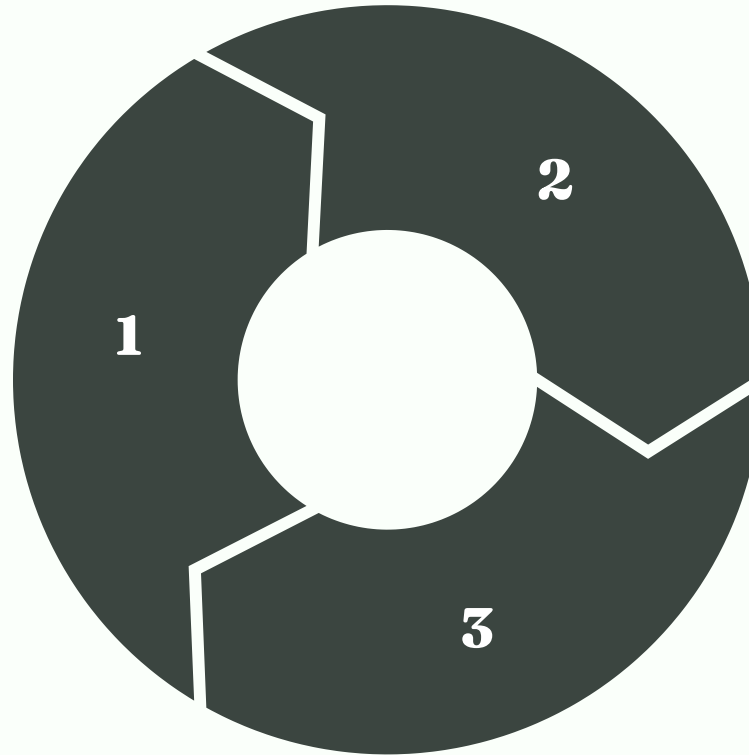
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Our Impact

Our 10-household demo has already offset 524 kilograms. Scale that to a suburb, a city, a country — and you've built a decentralized clean energy infrastructure that no utility can shut down



Every kilowatt traded on our network offsets 0.42 kilograms of CO2

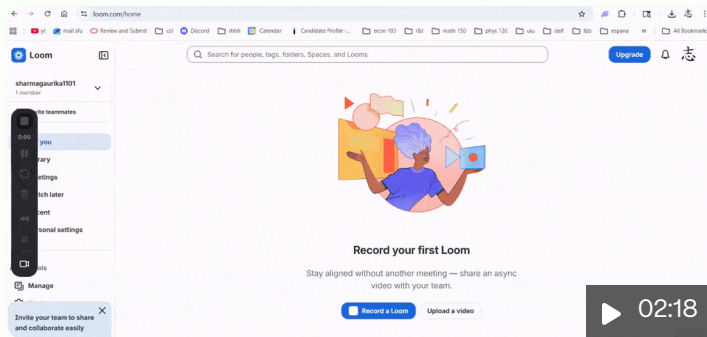
Directly addresses SDG 7 — affordable clean energy — and SDG 11 — sustainable cities



Thank You!

~ Praneet, Gaurika,
Tanveer, and Lucas

SDG: 7 & 11 — WattExchange



 Loom

For you | Loom – 6 April 2026

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